

to the therapist's assessment because the therapist can actively assess what is provoking pain. Concomitant facet arthritis, painful disc degeneration, and muscular pain can influence pain patterns and aggravating movements.

Adjunctive modalities can aid in relieving some associated symptoms. These modalities include ultrasound treatment, electrical stimulation, and massage. These may be helpful in short-term, symptomatic relief of back pain. Traction is also commonly prescribed. It theoretically may diminish intradiscal pressure, increase foraminal dimensions, and possibly relieve radicular pain secondary to herniated discs.⁵⁶ The role of chiropractic manipulation is controversial. Although some patients believe that manipulations have "reversed" the herniation, there is no evidence to support the ability of chiropractic manipulation to alter the normal or pathologic morphology of the disc.⁵⁷

Pharmacologic Treatment

Medications can be useful in decreasing disc-related symptoms. Because a local or systemic inflammatory reaction may participate in pain generation, antiinflammatory agents are believed to be beneficial. Nonsteroidal antiinflammatory drugs (NSAIDs) are first-line agents. Numerous choices are currently available, including cyclooxygenase-2 inhibitors. In prescribing NSAIDs, several important issues must be considered. The medications can have side effects. A history of gastrointestinal bleeding or peptic ulcers can be a contraindication to NSAID use. Although this risk is reduced with cyclooxygenase-2 agents, it is not nil, and the patient should be warned of the possibility. Additionally with NSAIDs, other systems, such as the kidneys and liver, may be detrimentally affected.

In the acute setting, back and radicular pain can be severe. Short-term narcotic use, such as a single dose of a morphine-derivative analgesic, can be useful. Narcotics should not be prescribed or administered in an extended manner; they should preferably be limited to a 2- to 3-day course. Also in the acute setting, a tapering dosage regimen of oral steroids can be helpful in decreasing inflammation-generated pain from nerve root irritation. In a randomized, double-blind, placebo-controlled trial of patients with acute sciatica due to herniated disc, Goldberg et al. demonstrated that a 15-day tapered dose of oral prednisone, compared with placebo, resulted in statistically significant, though modest, functional improvement without improvement in pain.⁵⁸

So-called muscle relaxants are frequently prescribed. Although the drug class name implies a direct muscular effect, these medications have more significant sedative effects. Medications such as diazepam and methocarbamol should be used sparingly. Truly antispasmodic medications such as baclofen or cyclobenzaprine can have a more direct effect on muscle spasms.

Selective transforaminal steroid injections can produce symptomatic relief in many patients. In our treatment protocol, injections are offered to patients who have failed noninvasive measures but either are not interested in or are not good candidates for discectomy. In patients with more than one

level of herniation, selective nerve root injection can be useful in determining the symptomatic level. That said, similar efficacy in pain relief between fluoroscopically guided epidural steroid injection and local anesthetic injection has been demonstrated in a recent double-blind, randomized controlled study.⁵⁹

In a prospective series touted as a natural history study, Bush and colleagues⁶⁰ reported the results of 159 patients with CT-confirmed disc herniations treated with epidural steroid injections. Although 91% avoided surgery, this underscores a glaring problem of so-called natural history studies in that an interventional treatment was used. Although its exact biochemical effects are still being elucidated, steroid injections may be effective in "avoiding" surgery. In an analysis of SPORT data, Radcliff et al. noted that a higher percentage of patients randomized to surgery who subsequently received an epidural steroid injection switched to the nonsurgical treatment group (41%) compared to patients randomized to have surgery who did not have an epidural steroid injection (12%).⁶¹ The authors concluded that this is confounded by the higher baseline desire to avoid surgery in the group of patients who received epidural steroid injections.

With TNF- α being considered a significant contributor to radicular symptoms associated with lumbar disc herniation,⁶² recent attention has been paid to delivering anti-TNF- α medications into the epidural space. Freeman et al conducted a multicenter, randomized, double-blind, placebo-controlled phase IIa clinical trial to examine the efficacy of etanercept, a TNF- α inhibitor, injected transforaminally to treat symptomatic disc herniation.⁶³ Compared to placebo, etanercept was found safe and effective in providing significant therapeutic relief for patients with sciatica. When compared to epidural dexamethasone injection, etanercept has shown similar clinical efficacy in back and leg pain reduction, with no significant difference between the two medications.⁶⁴ There has also been recent interest in the use of transforaminal epidural clonidine injections, for its potential to dual action as an antiinflammatory and analgesic,⁶⁵ though this remains controversial.

In a well-designed retrospective study, Wang and colleagues⁶⁶ studied 69 patients who failed nonoperative (or, more accurately, noninvasive) care and had requested surgery as treatment. Instead, each patient was advised to undergo one or more transforaminal steroid injections at the affected root level. Of patients, 77% had clinical resolution and had not undergone surgery at an average follow-up of 1.5 years. In agreement with other studies, clinical success was not related to disc size, percentage canal compromise, or degree of motor weakness. From these findings, selective nerve root steroid injections seem to produce at least short-term relief of radicular symptoms from a herniated disc.

Operative Versus Nonoperative Treatment

The results of operative and nonoperative treatment for symptomatic herniated discs have been compared in numerous studies. The Maine Lumbar Spine Study group published

1-year and 5-year results of an ongoing comparison of surgically and nonsurgically treated patients.^{1,67} More than 500 patients were included in the prospective observational study without stringent clinical or radiographic criteria except for disc-related sciatica treated with at least 2 weeks of nonoperative care within 2 months of onset. The decision to undergo surgery was determined on an individual basis and was not randomized. At 1-year follow-up, surgically treated patients were less symptomatic than patients in the nonoperative group, despite the former being *more* symptomatic at initial presentation. Relief of back or leg pain was reported by 71% of operated patients compared with 43% of the nonoperative group. High satisfaction levels and improved quality of life were documented for the operative group. For the workers' compensation group, there was no difference in time to return to work between operative and nonoperative groups. A criticism of the study is a substantial attrition rate, with 24% of patients unavailable for final follow-up.

In the 5-year outcome report, 70% of surgical patients reported back or leg pain improvement, whereas 56% of nonoperatively treated patients reported improvement.¹ As with the 1-year results, a similar percentage of patients were receiving workers' compensation benefits in both groups with no difference in return to work at final follow-up. Reoperations were performed in 20% of the operative patients; 16% of patients initially treated nonsurgically went on to operation. The authors noted that the benefits of surgery versus nonoperative treatment were greatest in the early part of the study, within the first 2 years, and that at final follow-up these advantages were less apparent.

Most recently, the Maine Lumbar Spine Study Group published their 10-year follow-up results.⁷ Of the eligible patients initially enrolled, data were available for 85% of patients treated surgically and 82% of patients treated nonoperatively. A significantly larger percentage of surgical patients reported relief of low back and leg pain than patients treated nonoperatively. Similarly, surgical patients exhibited better function and satisfaction compared with nonoperative patients. Nonetheless, improvement in dominant symptoms was reported for both treatment groups. Work and disability status were similar for both groups.

The 2006 SPORT investigation was designed to be a rigorous, randomized, prospective, controlled study.^{8,9,68} In its execution, there was a high rate of crossover between operative and nonoperative groups and a substantial portion of patients not willing to be randomized, resulting in a large observational group. The randomized and observational arms of the investigation reported improvement in bodily pain, physical function, and Oswestry Disability Index (ODI) scores regardless of intervention. In the randomized arm of the study, between-group differences favored surgical intervention, but these differences did not reach statistical significance in an intention-to-treat analysis. With the difficulties encountered with non-adherence to assigned treatments (i.e., crossover), the intention-to-treat analysis may not be reflective of the true outcomes. In an as-treated analysis, surgical treatment showed statistically superior results compared with nonoperative treatment. Similar results were found in the observational arm

of SPORT. These results have proven to be durable, as there has been little to no degradation of outcomes in either the nonoperative group or the operative group in 8-year follow-up reports.⁶⁹ With the analysis adjusted for the treatment received, operated patients still maintained clinically significant greater improvement in all primary outcomes compared to the nonoperative group, as was reported at earlier follow-up time points.

The classic work by Weber¹⁰ reported similar results to these findings. As briefly discussed in the prior section on natural history, Weber¹⁰ compared surgery versus nonoperative care in a randomized, prospective study. There were three study groups: one group "required surgery"; another showed no indications for surgery; and the third was the "undecided" group, in which it was unclear if surgery would be beneficial. Only the third group was randomly assigned to surgical or nonsurgical treatment.

At 1 year, good results were reported in 33% of the nonoperative group versus 66% in the surgical group; at 4 years, 51% versus 66%; and at 10 years, 55% versus 57%. These findings show that operative and nonoperative treatment outcomes seem to converge with time, being nearly the same at 10 years, and that the benefits of surgery are early. The 5- and 10-year results from the Maine study are similar, although the larger numbers of the Maine study achieved statistical significance. Both studies are flawed by patient selection bias. Weber randomly assigned only patients in the "unclear" group, whereas the Maine group had no specific criteria and included patients per individual physician investigators' usual practice.

Alaranta and colleagues⁷⁰ also prospectively compared operated versus nonoperated patients. In contrast to other similar studies, the nonoperative group was subdivided into cases with and without myelographic evidence of nerve root compression, whereas all operated patients had a positive myelogram. At 1-year follow-up, 91% of operated and 82% of nonoperatively treated patients with positive myelograms had improved pain levels. Only 51% of nonoperatively treated patients with negative myelograms had improvement. These data suggest that a distinctly worse natural history exists for patients with sciatica and no evidence of root compression. Although not specified by the investigators, this group probably represents sciatica from an extraspinal or non-disc-related origin. In a companion study published by the same authors, they further identified this group as having a high incidence of generalized pain (e.g., concomitant occipital headaches), more physically strenuous jobs, and lower pain thresholds.⁷¹ This study could possibly point to a preponderance of psychosocial and behavioral factors involved in the perpetuation of the symptom complex.

Operative Treatment

Indications

An absolute indication for lumbar discectomy is a progressive neurologic deficit. In this circumstance, operative intervention may be considered conservative care, provided that no medical contraindications exist. Progressive neurologic deficit

is most commonly associated with a cauda equina syndrome and is discussed in more detail later. The relative indications for discectomy vary among surgeons and patients. Discectomy, in its many shapes and forms, can produce symptomatic relief in appropriately selected patients. It is the surgeon's obligation to identify the patients in whom the anticipated benefits outweigh the attendant risks of surgery.

A prerequisite is radiologic identification of compressive pathology that is concordant with the patient's physical signs and symptoms. A patient with a large left L4–L5 level who presents with leg pain that radiates to the dorsum of the foot, weakness of toe dorsiflexion, decreased sensation in the first dorsal webspace, and a distinctly positive ipsilateral and contralateral SLR is an ideal candidate after failure of appropriate nonoperative treatment. In this “ideal” situation, the clinical and radiologic findings point toward compression of the L5 nerve root. To idealize the presentation further, the patient has a strong desire to return to work; is not involved in litigation, disability, or workers' compensation issues; and does not have any psychological issues. Important questions remain unanswered even in such cases, however, including optimal surgical timing, the method of disc excision, and the postoperative rehabilitation protocol.

Only a few patients match such textbook descriptions. Most patients lack one or more of the supportive diagnostic clues, making it more difficult to support the decision to operate. This situation does not represent a contraindication to surgery, however, because many published series show approximately 85% success rates in patient groups with lesser percentages of objective motor and sensory findings.^{3,38,67,72–79} The slim chance of back pain relief in relation to leg pain with discectomy that surgeons obligatorily confess to their patients is also probably an underestimation because most series document at least modest improvements in back pain. It is incumbent on clinicians to discuss the advantages, disadvantages, risks, alternatives, and estimated expected outcomes with patients.

Available Techniques

A vast array of techniques exist for surgical treatment of herniated discs.^{80–85} Standard open discectomy is the most common surgical approach.^{72,77,86,87} It involves careful incision planning, laminotomy or partial laminectomy to provide adequate visualization of the pathology, gentle retraction of the neural elements, and direct excision of the herniation. As an adjunct to open discectomy, some surgeons advocate the use of a microscope for better visualization and minimizing incision size.^{31,79,83,88,89} The purported advantage of the microscope is the ability for the surgeon and the assistant to visualize the operative field equally through a smaller surgical wound.

Alternatives to interlaminar techniques have been developed for excision of foraminal and extraforaminal lateral disc herniations, which involve exposures between the transverse processes and lateral to the pars interarticularis.^{14,89} Although some surgeons have advocated the addition of fusion,^{76,90} this practice is unpopular.⁸¹ Advocates hold that fusion decreases

the chance for reherniation; however, only a complete discectomy with interbody reconstruction can eliminate this risk. Long-term effects such as an adjacent-segment degeneration and the additional morbidity and complication rates are potential disadvantages.^{76,81,90,91}

With increased interest in smaller incisions and minimally invasive surgery, various percutaneous methods of treatment have been developed. Some methods entail placement of a cutting device intradisally to decompress the disc space to retract the herniated fragment.^{92–95} Other methods involve percutaneous techniques of directly visualizing the neural elements and disc using an endoscope.^{80,96} Chemical digestion of the disc (i.e., chemonucleolysis) had enjoyed popularity in the past. However, enzyme-related complications and results inferior to open discectomy limited continued popularity in the United States.^{92,97,98}

Repair of the annulus fibrosus following lumbar discectomy is an emerging technology, with various techniques and devices introduced in recent years. There remains a paucity of published studies concerning the clinical outcomes of newer devices, though data are forthcoming. In one published prospective randomized trial, an annular repair technique did not show statistically significant benefit in patient symptoms, but did reduce the need for subsequent surgery.⁹⁹

Open Simple Discectomy

Timing

Disagreement exists regarding the optimal timing for surgery. The question is how long should nonoperative care be continued (without improvement) before the outcomes of surgery are detrimentally affected? Stated another way, in patients who are not responding to nonoperative measures, when should surgery be performed? In evaluating the currently available surgical results, this interval may be 2 to 16 months. The clear exception to delaying surgery is a patient who is experiencing a progressive neurologic deficit or cauda equina syndrome.

Rotheorl and colleagues³³ stratified operatively treated patients according to time from presentation to surgery. Patients with symptom duration of more than 2 months had a statistically significantly worse outcome than patients operated on within 2 months. There was no difference if surgery was performed within 1 or 2 months. Likewise, Hurme and Alaranta¹² found the best results in patients operated on within 2 months of the onset of disabling sciatica. Nygaard and colleagues³² reported worse results in patients with leg pain for 8 months or more; Jansson and colleagues³⁶ found similar results in their analysis of the Swedish registry data, which included greater than 27,000 patients. Sorensen and colleagues³⁴ found that symptom duration greater than 16 months was predictive of poor results, but this was highly influenced by patient personality and social factors. The disparity between these findings is difficult to explain. Randomized prospective data comparing early versus late surgery are lacking. In our practice, surgery is performed within 6 months from symptom onset if nonoperative care has failed. The important message